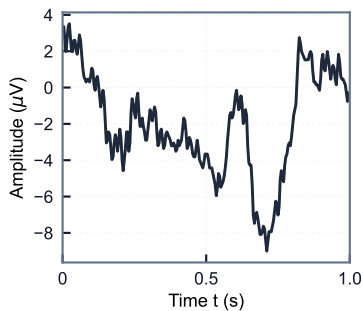
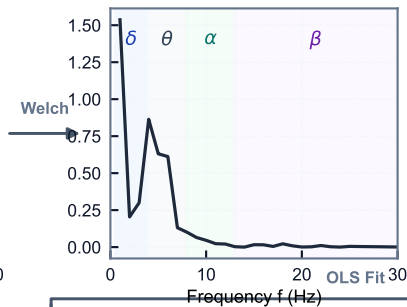
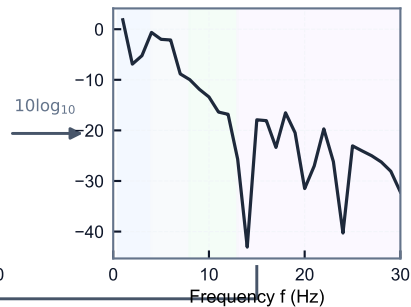
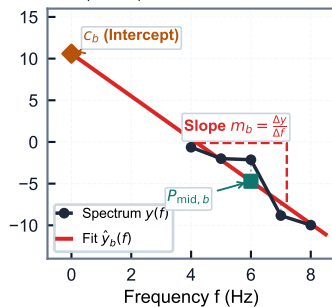


(a) Raw EEG Epoch $x(t)$ **(b) Welch PSD $P(f)$** PSD ($\mu\text{V}^2/\text{Hz}$)**(c) Log-PSD Dynamics $y(f)$** $10\log_{10} P(f)$ (dB/Hz)**(d) Sub-Band OLS Fit $\hat{y}_b(f)$**

Power (dB/Hz)

**(e) Biomarker Vector $\mathbf{x}_{\text{slope}}$** **1. Sub-Band Vector:**

$$\mathbf{f}_b = [m_b, c_b, p_{\text{mid},b}]^T$$

2. Channel Vector:

$$\mathbf{f}_c = [\mathbf{f}_\delta^T, \mathbf{f}_\theta^T, \mathbf{f}_\alpha^T, \mathbf{f}_\beta^T]^T$$

3. Channel Dim:

$$\dim(\mathbf{f}_c) = 4 \times 3 = 12\text{D}$$

4. Multichannel:

$$\mathbf{x}_{\text{slope}} = [\mathbf{f}_1^T, \dots, \mathbf{f}_C^T]^T$$

$$\mathbf{x}_{\text{slope}} \in \mathbb{R}^{C \times 12}$$

$$(C = 18 \rightarrow 216\text{D})$$

Concatenate